abstracted away). *Message queuing telemetry transport (MQTT)* is an open source, lightweight messaging protocol that has become the default application protocol for IoT. MQTT uses a loosely coupled, publish/subscribe communication model, which is highly flexible and adaptable and decouples devices and applications.

I use MQTT to transport data from all the sensors that I am integrating to a digital twin application server, which serves as a collection hub for all incoming data. It is not only data from networked electronic sensors that can be transported over MQTT—I also use it to transport manually collected data into my digital twin.

Whatever the source of data, the transformation to MQTT involves several steps:

1. *Protocol translation*—convert from sensor-specific protocols to MQTT messages. The managed network server from AWS handles this conversion to MQTT for LoRaWAN sensors.
2. *Data organization*—assign meaningful hierarchies for data organization.
3. *Payload standardization*—transform diverse formats into consistent structures.
4. *Metadata preservation*—maintain context about data origin and quality.

MQTT IN PRACTICE

Traditional point-to-point communication creates rigid dependencies between systems. If a temperature sensor needs to send data to a monitoring dashboard, a database, and an alerting system, it must know about and maintain connections to all three consumers. When requirements change, such as adding a new analytics service or removing an old system, the sensor’s code must be modified. This tight coupling makes it difficult to extend and evolve such communication systems.